

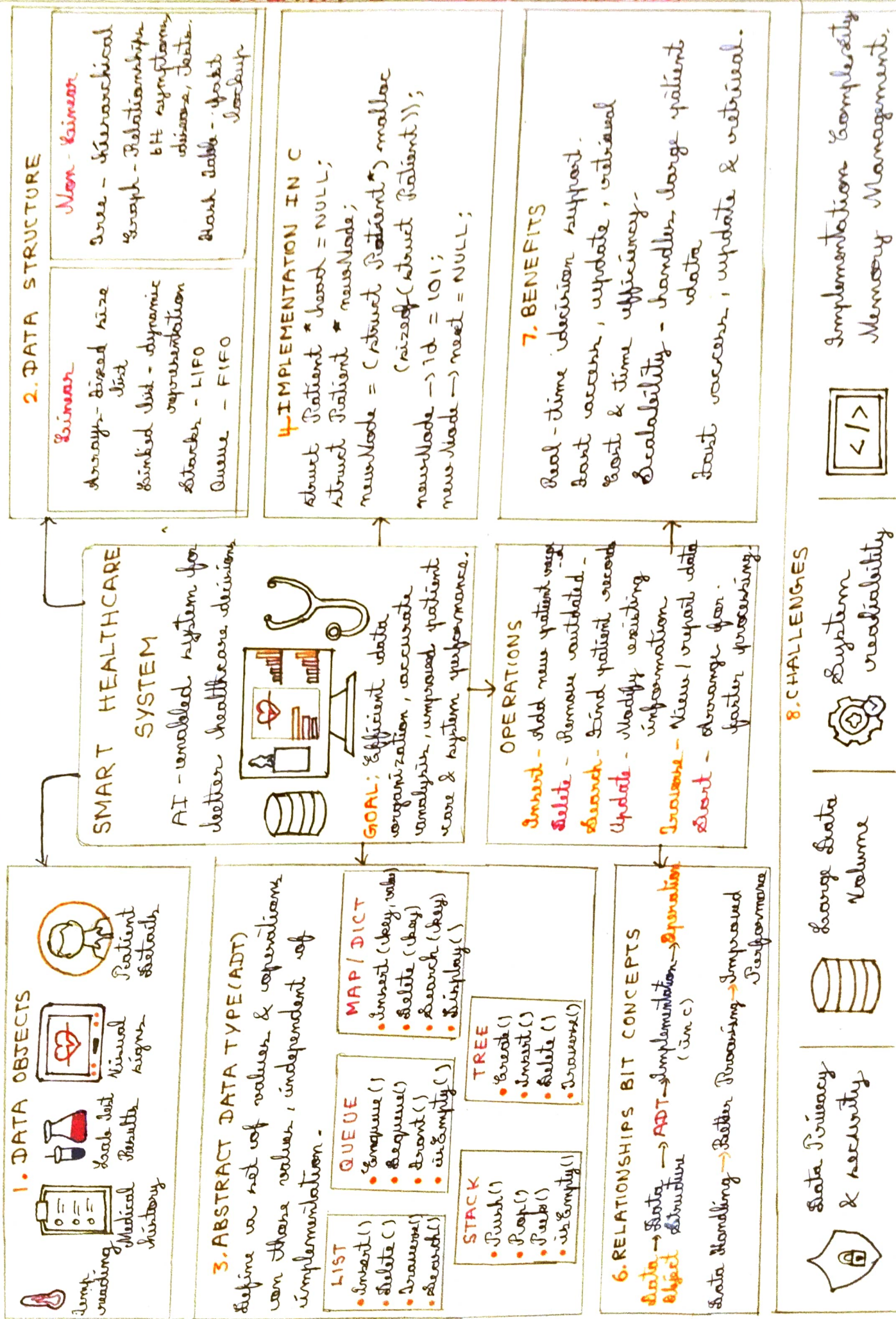
Q1: SMART HEALTHCARE SYSTEM

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COURSE CODE: CSA0313

COURSE NAME: DATA STRUCTURES

CO1: AT-2 - CONCEPT MAPPING

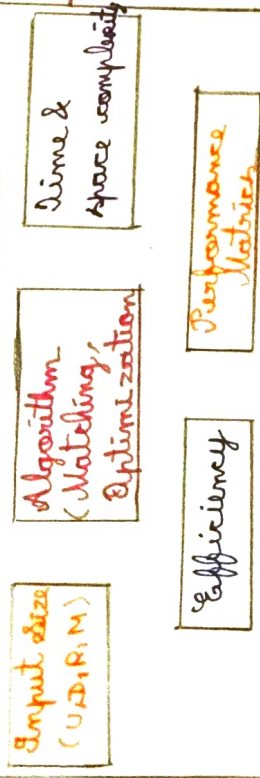


Q4: RIDE - SHARING APPLICATION

1. ALGORITHMS

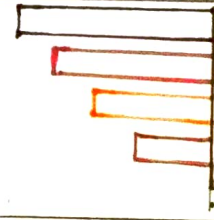
- Matching Algorithm - Find nearest driver (using distance, ratings)
- Dynamic Pricing Algorithm
- Route Optimization
- Real-time Update Algorithm

7. RELATIONSHIP FLOW

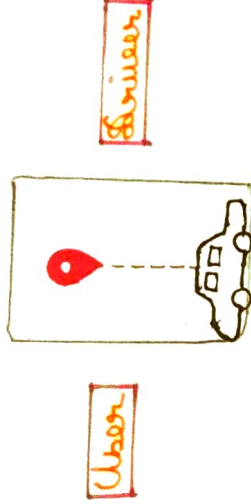


6. PERFORMANCE MEASUREMENT

Average Match Time
Request fulfillment Rate
Driver Utilization
User Satisfaction rating
System throughput.



RIDE - SHARING APPLICATION



5. EFFICIENCY

- ✓ Use spatial index to reduce search time.
- ✓ Optimize routes - less fuel & time.
- ✓ Efficient memory usage.
- ✓ Real-time responsiveness.



2. INPUT SIZE

- Number of Users (U)
- No. of Drivers (D)
- No. of location Requests (R)
- Map Data Size
- Attributes: Rating, ETA, Distance.



3. TIME COMPLEXITY

- Naive Search (all drivers)
- Spatial Index (KD-Tree)
 $R\text{-Tree}: O(\log D)$
- Matching for all requests:
 $O(R \log D)$
- Route Computation (Shortest Path): $O(E \log V)$



4. SPACE COMPLEXITY

- Store Users: $O(U)$
- Store Drivers: $O(D)$
- Map Graph (V Vertices, E edges):
 $O(V + E)$
- Auxiliary structures (Indices, Queues):
 $O(D \log D)$

